Data Mining Practice Question
Set 2

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1. Consider the following dataset of students' study habits and exam results where each student passed or failed the exam based on their study hours and attendance.

Student	Study Hours (H)	Attendance (A)	Class (Y)
S1	High	Good	Pass
S2	Medium	Average	Pass
S3	Low	Good	Fail
S4	Medium	Good	Pass
S5	Low	Average	Fail
S6	Medium	Poor	Fail
S7	High	Average	Pass
S8	Low	Poor	Fail
S9	High	Good	Pass
S10	Medium	Average	Pass
S11	**Medium**	**Good**	**?**
S12	**High**	**Poor**	**?**

a) Compute all the conditional probabilities for each feature given each class.
b) Using the Naïve Bayes classification, predict the class for a new student S11.
c) For new student S12, determine whether the Naïve Bayes classifier can make a valid prediction. If not, identify the reason and propose an appropriate remedial technique.

d) Critically analyze the assumption of conditional independence in Naïve Bayes. Discuss how correlation between Study Hours and Attendance (for example, poor attendance often leading to low study hours) could affect model performance.

2. What is regression? Explain graphically, the overfit and underfit regression models on a sample of data. Also show the best fit model.

   For the following data,

X	2	3	5	6	8
Y-True	6	4	5	8	10

   Derive the Mean Squared Loss Function for the linear regression function and calculate the training MSE.

3. Consider the dataset

``Student	**Study Hours (SH)**	**Mode of Study (MS)**	**Attendance (AT)**	**Marks (MS)**	**Result (R)**
S1	High	Offline	High	Good	Pass
S2	High	Online	Low	Good	Pass
S3	Medium	Online	High	Average	Pass
S4	Low	Online	High	Average	Fail
S5	Low	Online	Low	Poor	Fail
S6	Medium	Offline	Low	Poor	Fail

   Given that the attribute Result is the class label, construct the complete decision tree using the Information Gain as attribute selection method.

4. A company wants to predict whether a candidate will be Hired or not based on three binary features. The training dataset of the candidates is as follows:

Candidate	Degree	Experience	Skillset	Hired
1	Yes	Yes	Yes	Yes
2	Yes	Yes	No	Yes
3	No	Yes	Yes	No
4	No	No	Yes	No
5	No	Yes	No	No
6	Yes	Yes	Yes	Yes
7	No	No	No	No
8	Yes	No	No	Yes

Using Simple Naive Bayes, determine whether the new candidate who **has degree**, **do not have experience** but **have skillset** will be hired or not.